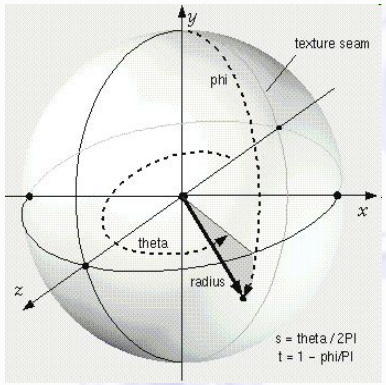
7. Illumination

Phong Illumination
Diffuse, Specular and Ambient
Attenuation, Positional sources

Rendering

- Recall the two key-processes in graphics

CONCEPT



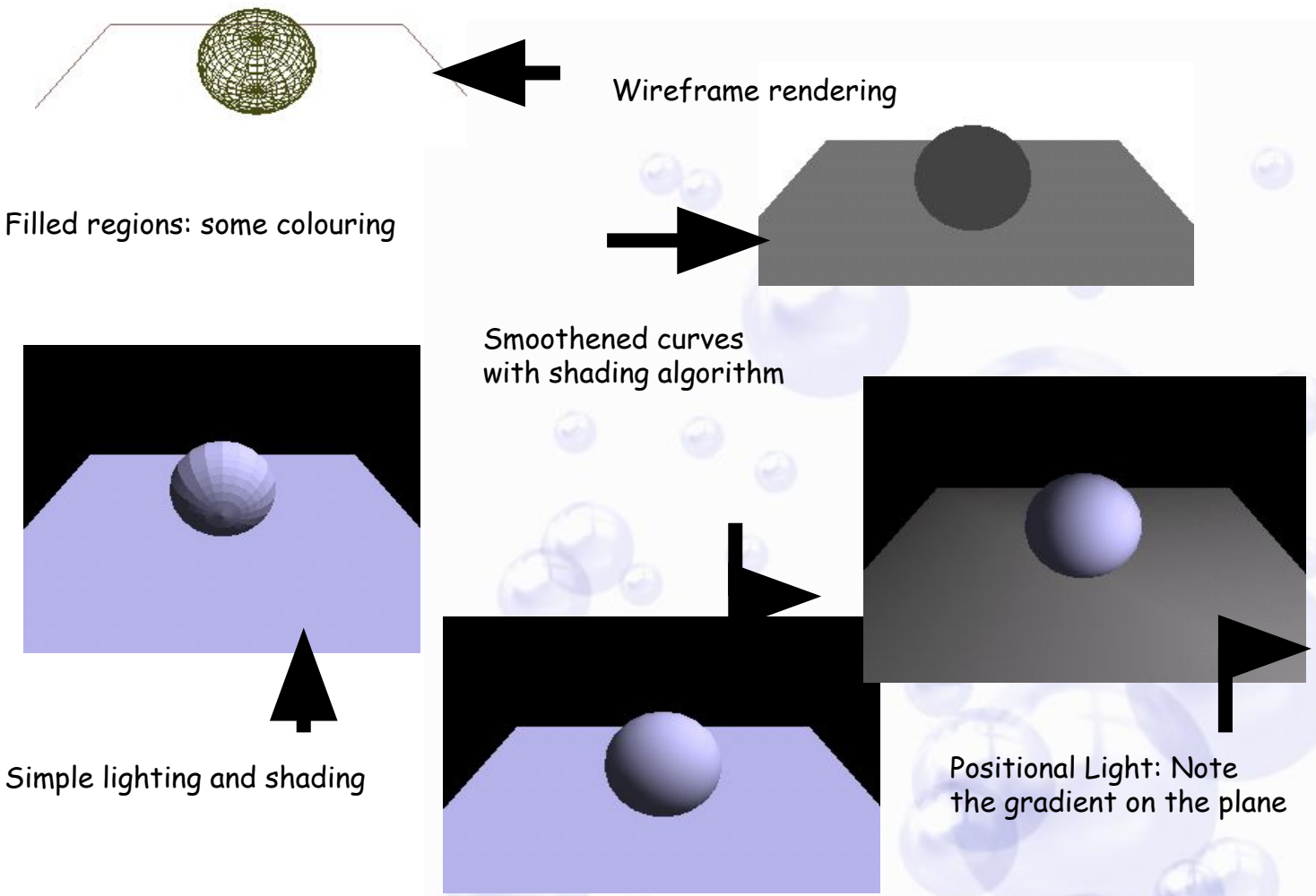
MODEL

IMAGE

modelling

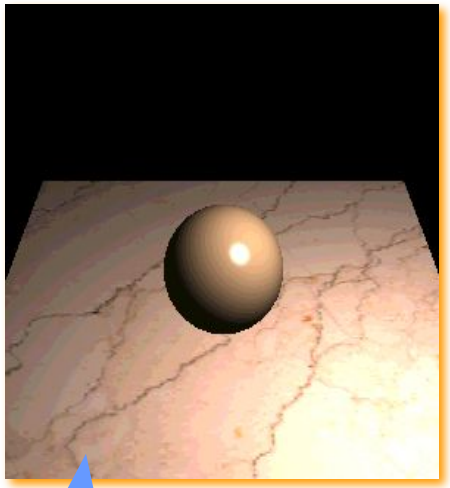
rendering

- We will now address the 3D rendering problem in detail.



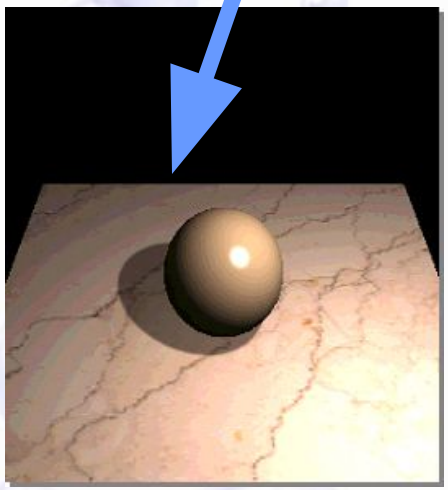
The Same Model can be rendered in many different ways. No rendering style is necessarily less-correct and maybe ideal for a specific application.

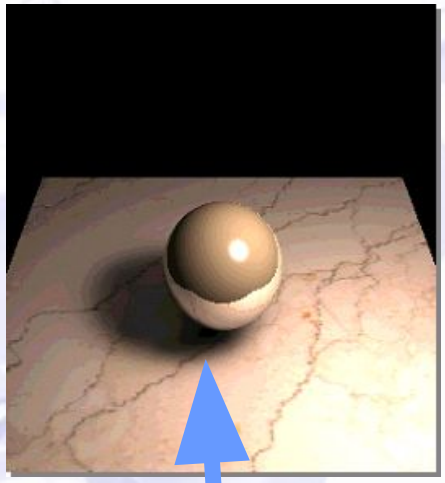
Further different types of rendering.



A bit of texturing enhances the scene considerably making it look more *"real-world-like"*

Fake shadow: gives a better idea of what the image represents (i.e. position of sphere is more apparent)





Global Illumination: "*proper*" shadows, specular reflections on objects

Light and Shadow

When an artist renders a 3D scene he tries to accurately portray the interaction of light and shadow on a scene. In Computer Graphics we try to achieve a similar portrayal using precisely defined mathematical/algorithmic steps.

Rendering

- fundamentally concerned with determining the most appropriate colour (i.e. RGB triple) to assign to a pixel associated with an object in a scene.

- The colour of an object at a point depends on:
  - geometry of the object at that point (normal direction)
  - position, geometry and colour of the light sources (luminaires)
  - position and visual response of the viewer
  - surface reflectance properties of the object at that point
  - scattering by any participating media (e.g. smoke, rising hot air)

Lighting a Scene

- All surfaces considered to contribute by emitting light or reflecting
- The color of any point in the scene is determined by multiple interactions among light sources and reflective surfaces
- Recursive process of light transfer causes subtle effects such as colour bleeding between adjacent surfaces
- Mathematically represented as an integral equation

Rendering Algorithms

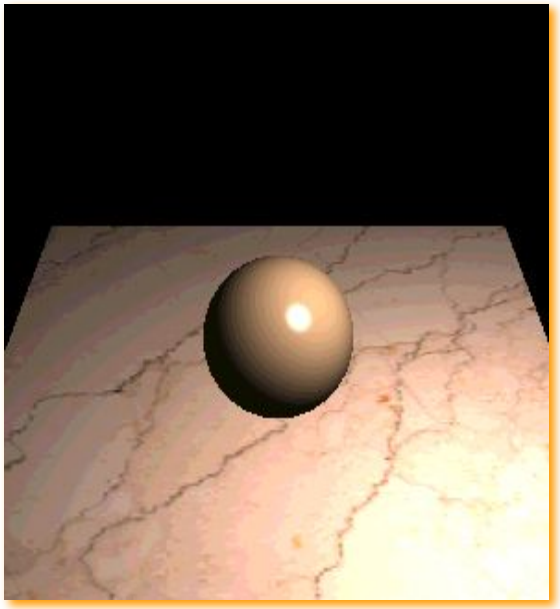
- To make this problem solvable in finite time certain assumptions/simplifications need to be made.

- Rendering algorithms differ in the assumptions made regarding lighting and reflectance in the scene and in the solution space:

  - **local illumination** algorithms: consider lighting only from the light sources and ignore the effects of other objects in the scene (i.e. reflection off other objects or shadowing)
  - **global illumination** algorithms: account for all modes of light transport

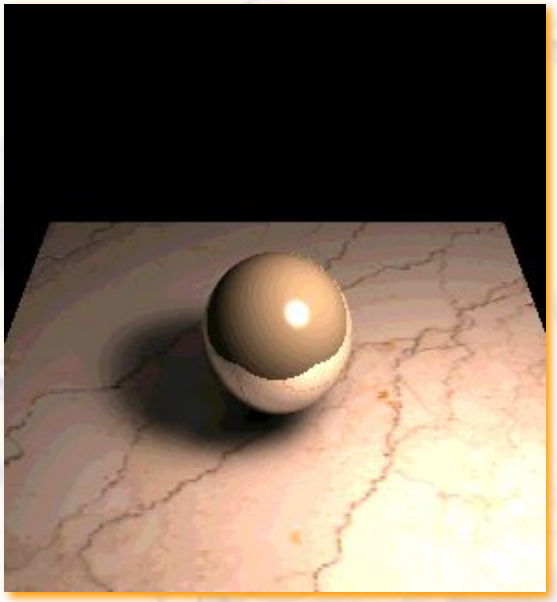
  - **view dependent** solutions: determine an image by solving the illumination that arrives through the viewport only.
  - **view independent** solutions: determine the lighting distribution in an entire scene regardless of viewing position. Views are then taken after lighting simulation by sampling the full solution to determine the view through the viewport.

Local vs. Global Illumination



Local

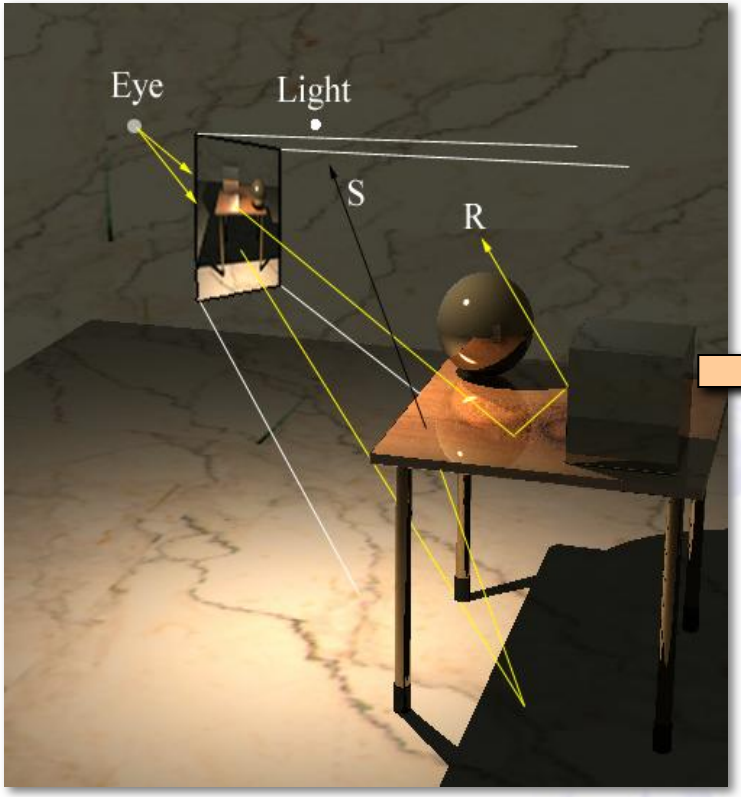
Illumination depends on local object &
light sources only



Global

Illumination at a point can depend
on any other point in the scene

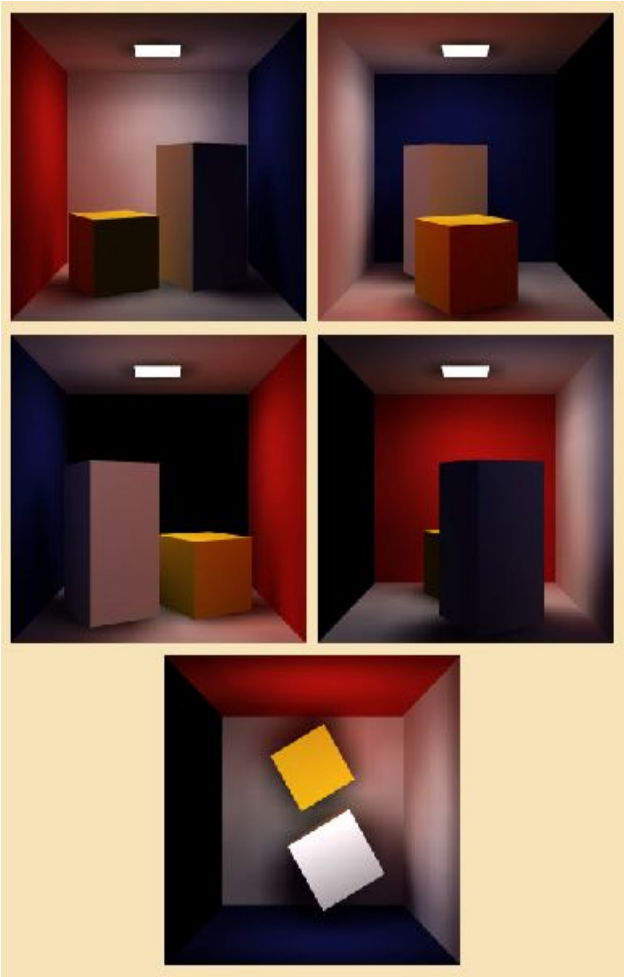
A View Dependent Solution (Ray-tracing)



Scene Geometry

Solution determined only for directions through pixels in the viewport

A View Independent Solution (Radiosity)



A single solution for the light distribution in the entire scene is determined in advance.

Then we can take different snapshots of the solution from different viewpoints by sampling the complete solution for specific positions and directions.

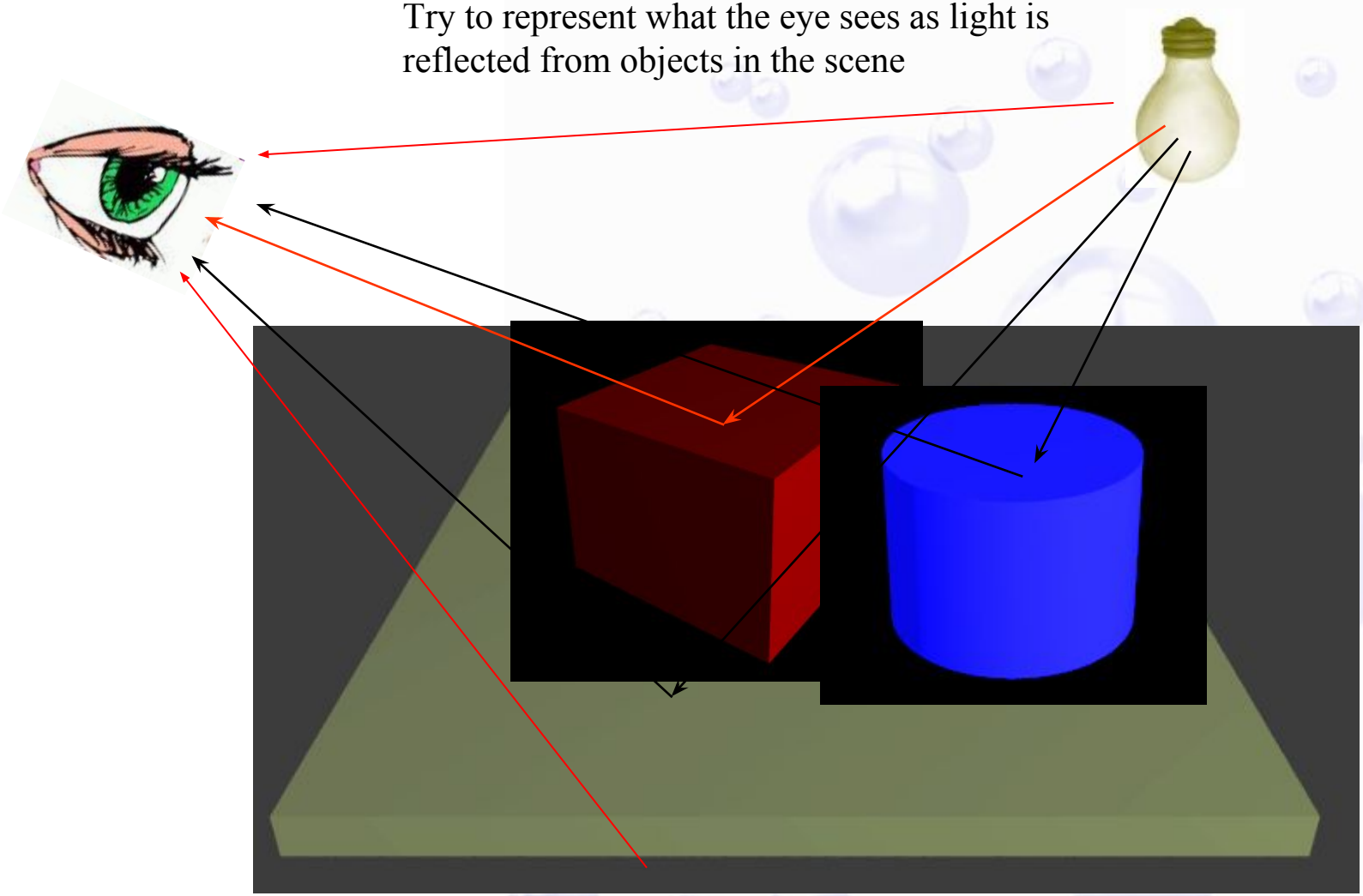
Illumination Model

- Lighting is described with models that consider the interaction of electromagnetic energy with object surfaces

- An **illumination model** is used to calculate the intensity of light that we see at a given point on the surface of an object in a specified viewing direction
- Illumination models are derived from physical laws that describe surface light intensities
- A surface-rendering algorithm uses the intensity calculations from an illumination model to determine the light intensity for all projected pixel positions for the various surfaces in a scene.

Illumination Models

Try to represent what the eye sees as light is reflected from objects in the scene



Illumination Variables

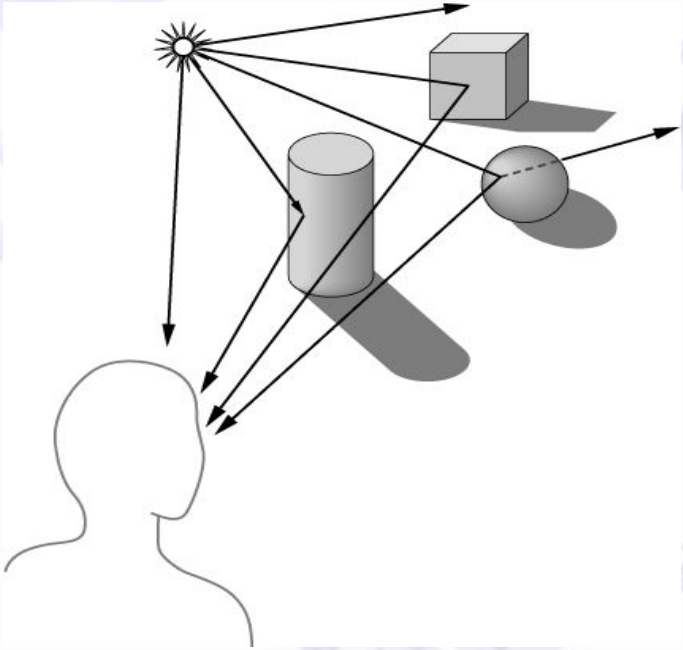
- Light source
  - Positions
  - Properties

- Object
  - position relative to lights
  - position relative to other objects
  - material properties
    - opaque/ transparent, shiny/ dull, texture surface patterns

- Position and orientation of view plane

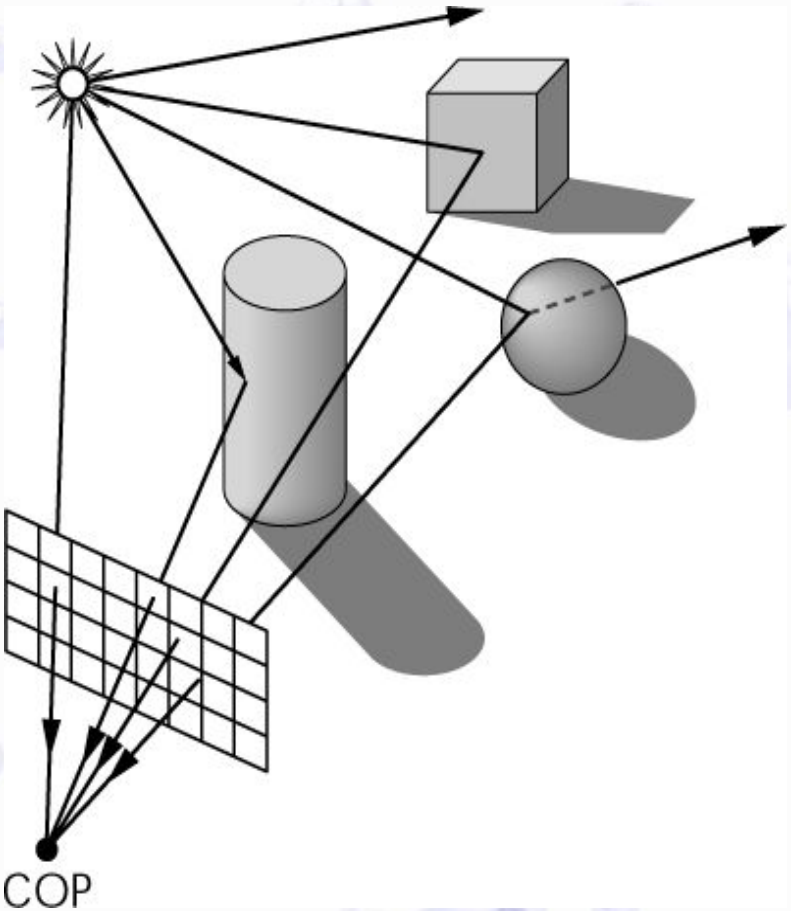
A Model for Lighting

- Follow rays from light source
- only light that reaches the viewers eye is ever seen
  - direct light is seen as the colour of the light source
  - indirect light depends on interaction properties



Lighting in Computer Graphics

- For Computer graphics we replace viewer with projection plane
- Rays which reach COP after passing through viewing plane are actually seen
- Colour of pixels is determined by our interaction model



Interaction Between Light and Materials

- The nature of interaction is determined by the material property
- Colour, smoothness and brightness of an object is determined by these interactions

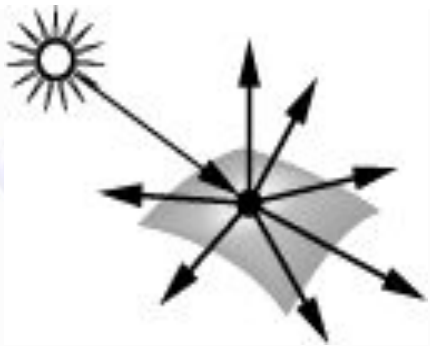
- Light hitting a surface is either absorbed, reflected or transmitted through material to interact with other objects

- Shading also depends on the orientation of the surface
- Three general types of Light-Material Interactions
  - Diffuse scattering
  - Specular reflection
  - Transmission / Refraction

Specular Surfaces

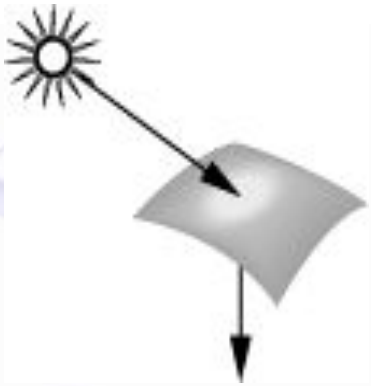
- Surfaces appear shiny
- Reflected light is scattered in a narrow range of angles close to angle of reflection
- Mirrors are perfectly specular surfaces: all light is reflected in a single direction (direction of perfect reflection)

Diffuse Surfaces



- Rough “matte” surfaces scatter incoming light
- Characterized by:
  - light scattered in all directions
  - end up appearing to have consistent “chalky” texture
  - perfectly diffuse surfaces scatter light equally in all directions. Also called ideal diffuse reflectors or Lambertian reflectors. Since radiated light energy from any point on the surface is governed by Lambert’s cosine law.

Translucent Surfaces



- Properties:
  - allow some light to penetrate and emerge from another location
  - an accurate model possibly involves refraction
  - some incident light may also be reflected at the surface

Phong Illumination

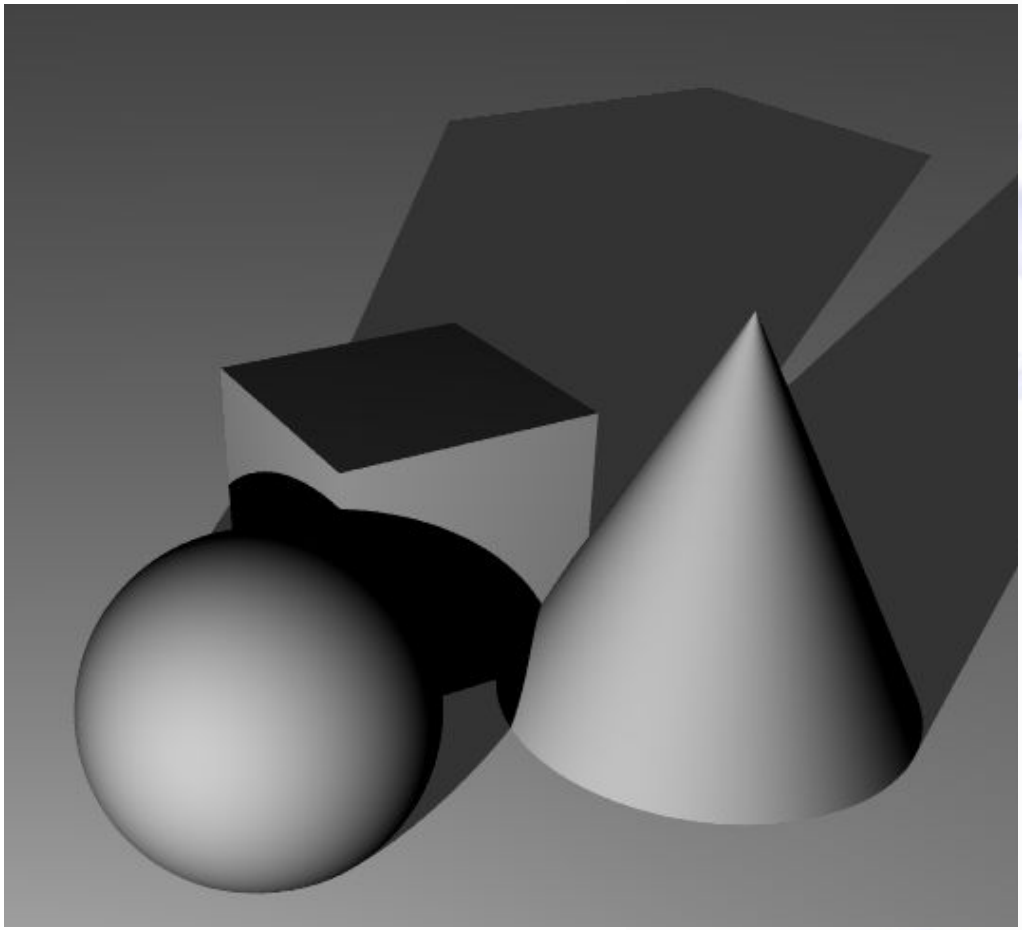
A local illumination model attributed to Bui Tong Phong
University of Utah 1973

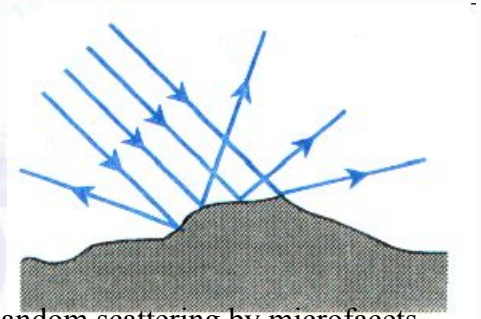
Model Assumptions

- In the following slides we assume
  - A point light source –
    - Position defined by a point in space, radiating light equally in all directions
    - Repeat and accumulate results if we wish to model more than one light source
  - A viewer
    - Position defined by a point in space, the centre of projection or camera positions

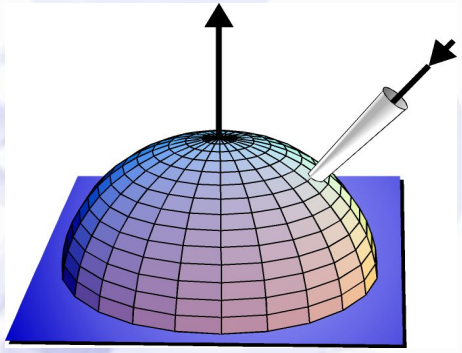
- In addition the equations in the following slides apply to monochromatic light (just intensity i.e. greyscale) for a colour solution, we need to repeat the steps for Red, Green and Blue components.

Diffuse Reflection



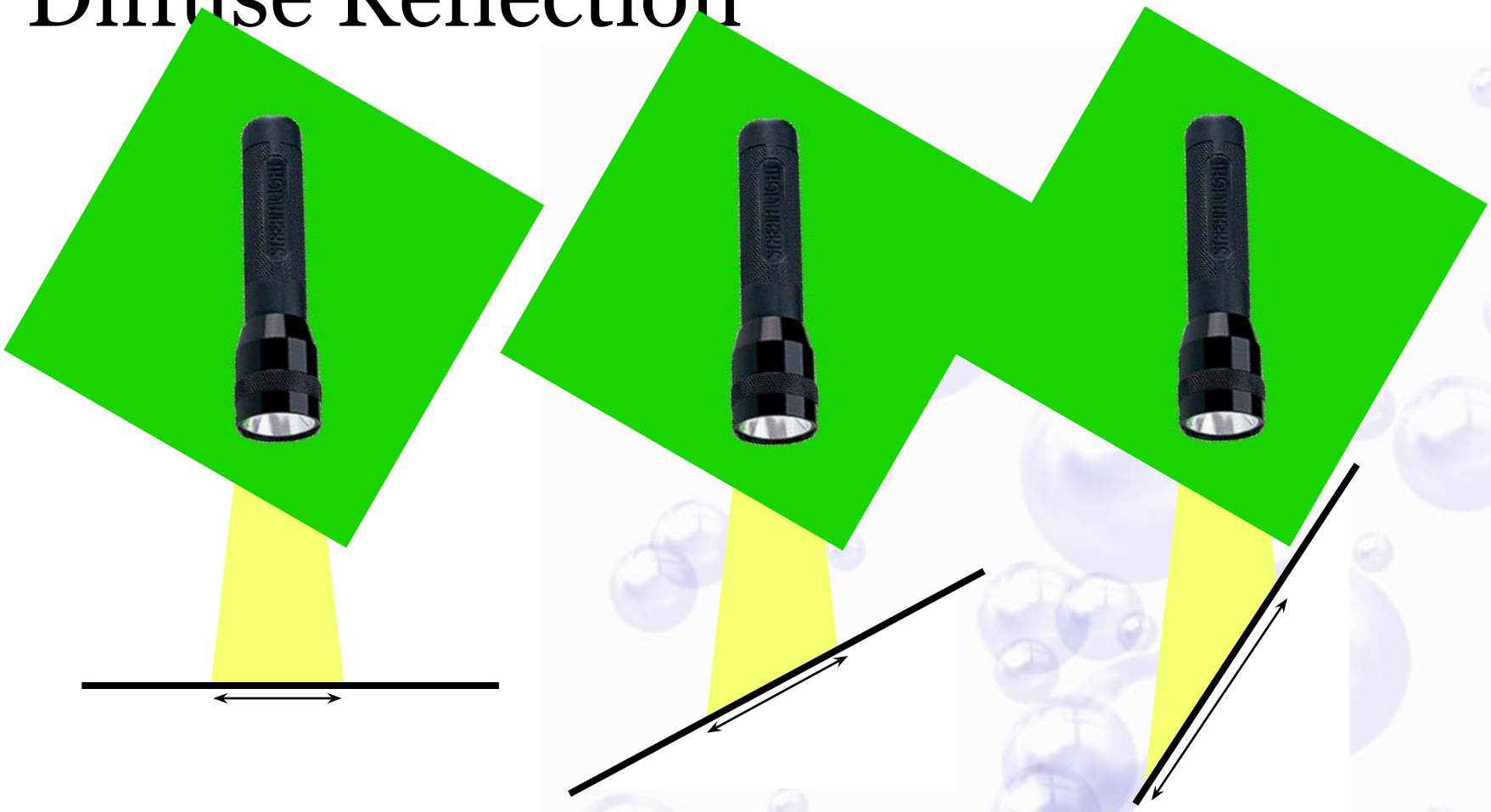


Random scattering by microfacets



Light from a point is invariant with viewing direction

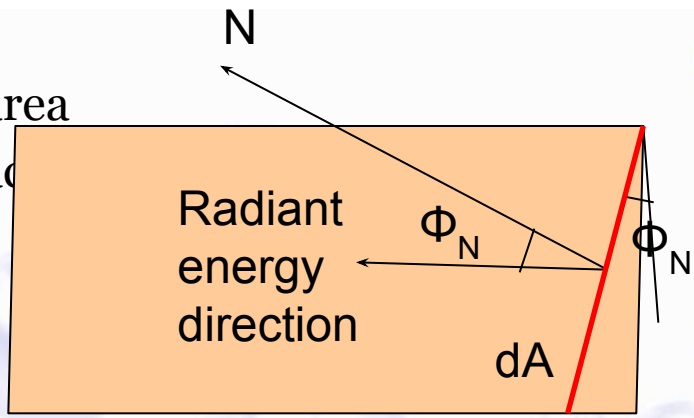
Diffuse Reflection



However the intensity of light reflected **IS** dependent on light direction

Lamberts cosine Law

- Radiant energy from any small surface area *dA* in any direction Φ_N relative to the surfac Normal is proportional to cos Φ_N.



The light energy depends on the radiant energy per projected area perpendicular to direction Φ_N*, which is* dA cos Φ_N.

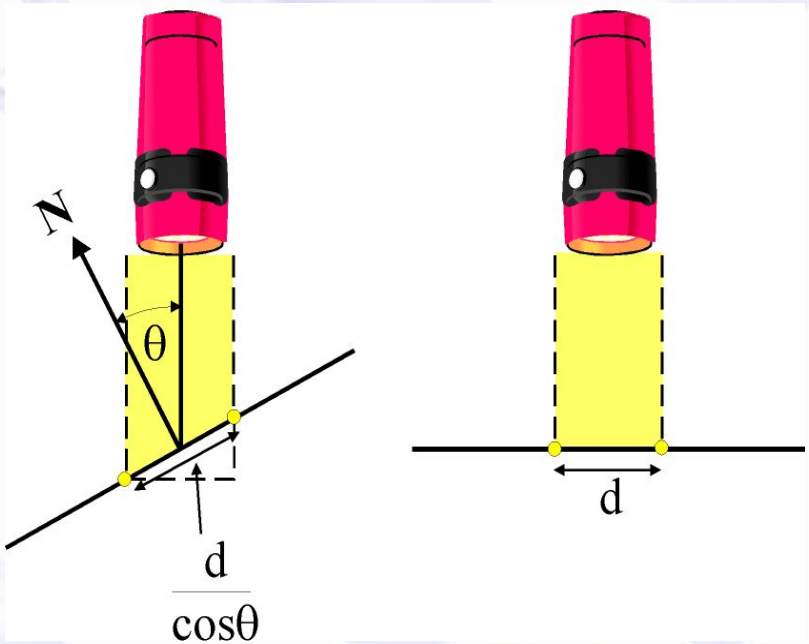
Thus for Lambertian reflection, light intensity is same over all viewing directions.

Lamberts Law

- A surface which is oriented perpendicular to a light source will receive more energy (and thus appear brighter) than a surface oriented at an angle to the light source.
- The irradiance E is proportional to $\frac{1}{area}$
- As the area increase the irradiance decreases therefore:

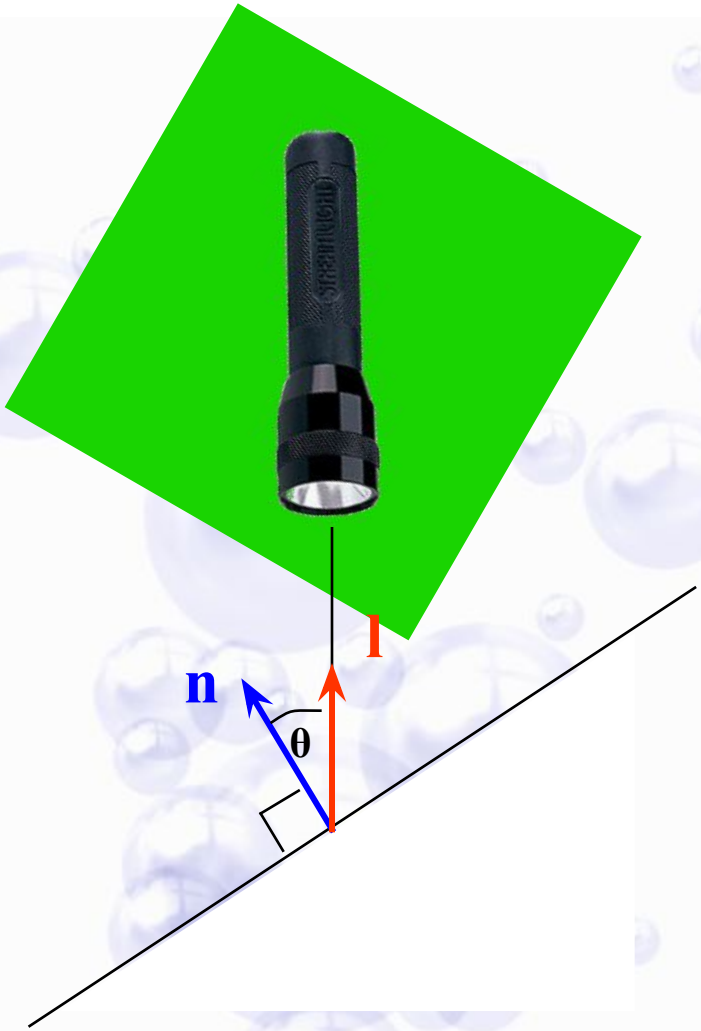
$$E = \frac{d\Phi}{dA_{\perp}} = \frac{\cos\theta\, d\Phi}{dA} = \frac{\cos\theta\, \Phi}{4\pi r^2}$$

As θ increases, the irradiance and thus the brightness of a surface decreases by cos θ



Lambertian Illumination Model

- Intensity of reflected light depends on:
  - The angle the light rays make with the surface of the object $\boldsymbol{\theta}$
  - And the reflectivity (*"colour"*) of the object surface ($\boldsymbol{k_d}$)
  - The original colour of the light ($\boldsymbol{L}$)



Graphical programs calculate light at each point using a simple formula

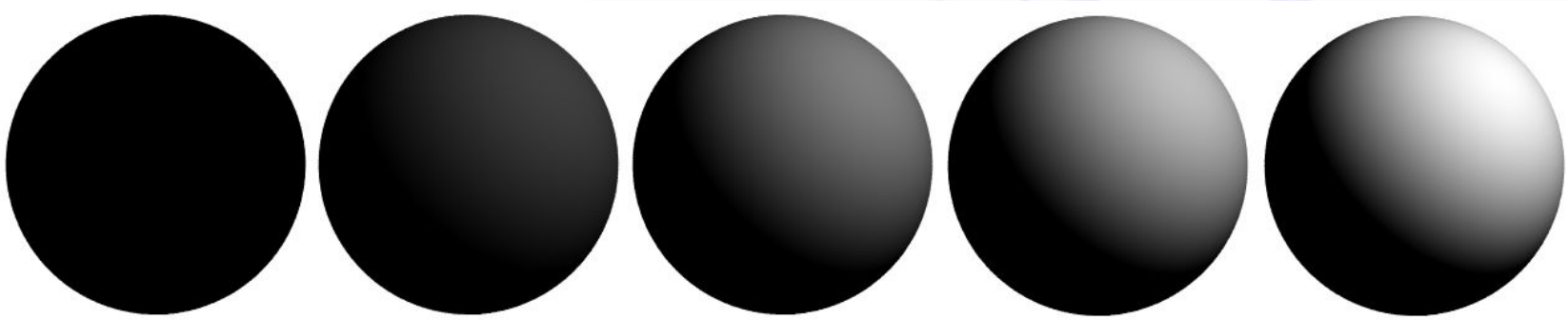
$$I_d = L \times k_d \times \cos\theta$$

θ is the angle between the normal and the light direction.

so $$I_d = L \times k_d \times (\mathbf{l} \bullet \mathbf{n})$$

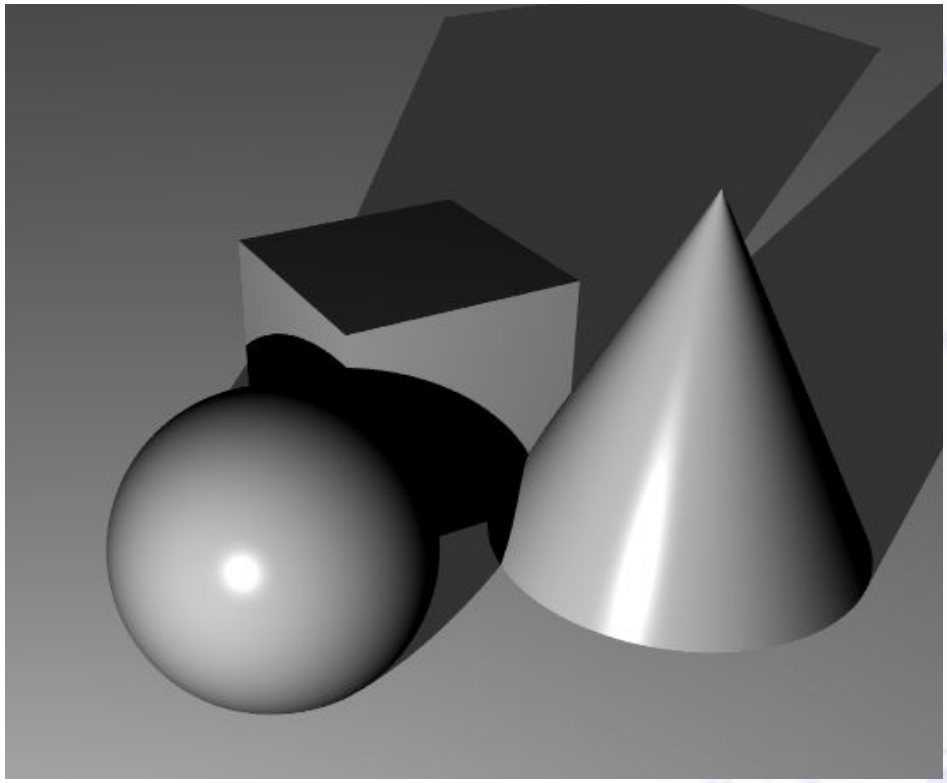
Where **l** and **n** are unit vectors

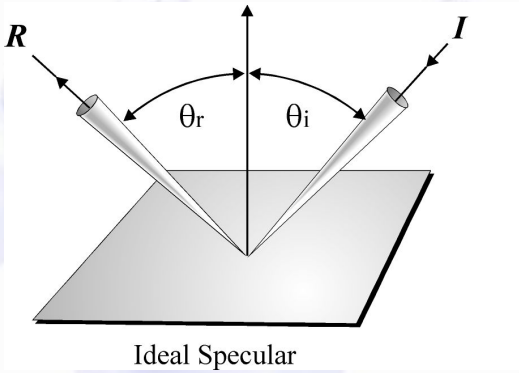
Diffuse Reflection



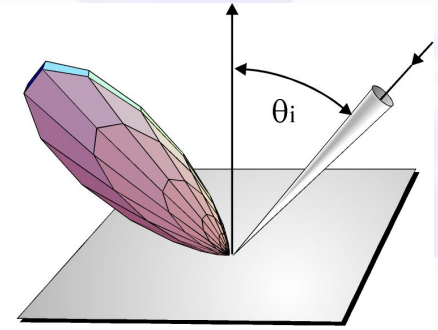
- The spheres above are lit by diffuse (k_d) values of 0.0, 0.25, 0.5, 0.75, 1 respectively

Specular Highlights





Ideal Specular



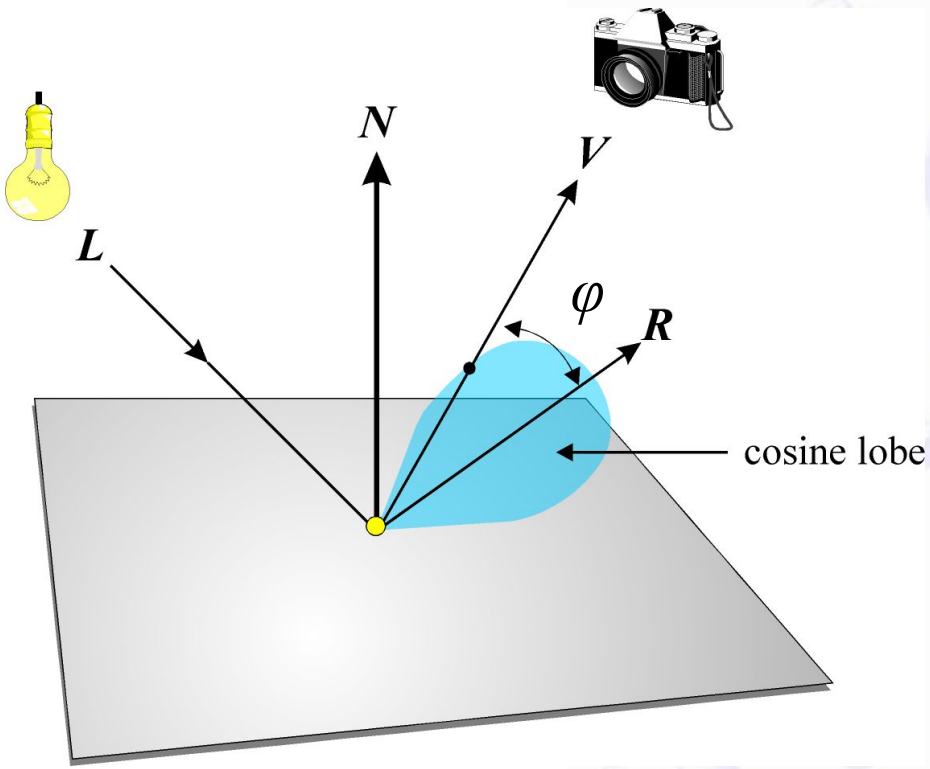
Rough Specular

Phong Illumination Model

- To simulate reflection we should examine surfaces in the reflected direction to determine incoming light
  - global illumination

- The Phong model is an *empirical* local model of shiny surfaces – A local model used to simulate effects which can be global in nature
- We only consider reflections of light sources. Assume that the intensity of shiny surfaces may be approximated by a spherical cosine function raised to a power (known as the Phong exponent).
- A useful approximation for efficient computation of light-material interactions which produces good renderings under a variety of lighting conditions and material properties

Phong Model of Specular Reflection



- Intensity reflected light depends on:
  - Viewer direction
  - Incoming light direction
  - Light colour + brightness (L)
  - Shininess/polish of material (α)
  - Reflectivity of material (k_s)

$$I_s = L \times k_s \times \cos^{\alpha} \phi$$

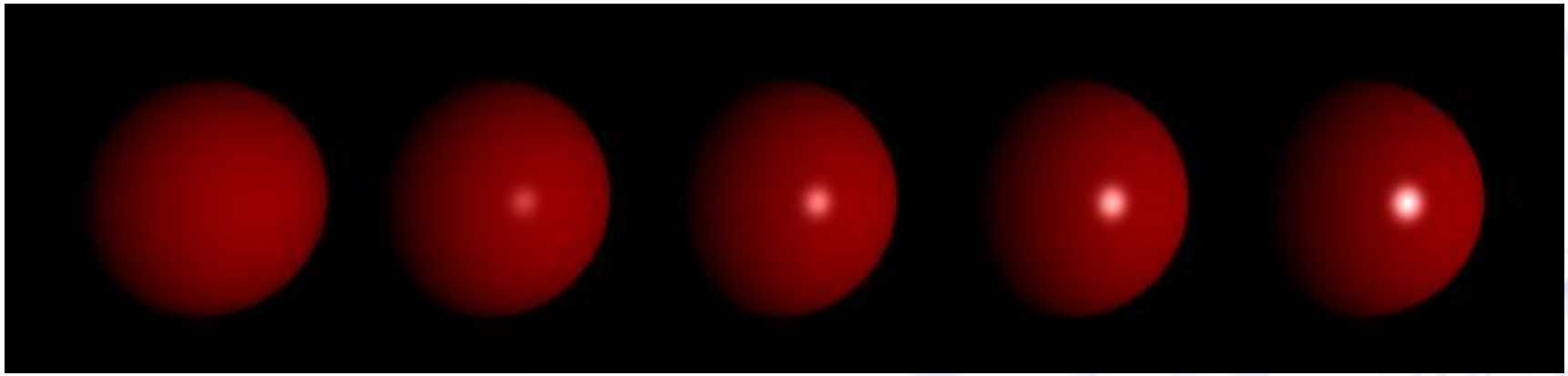
φ is the angle between the viewer and the reflected light direction.

OR $$I_s = L \times k_s \times (\mathbf{v} \bullet \mathbf{r})^{\alpha}$$

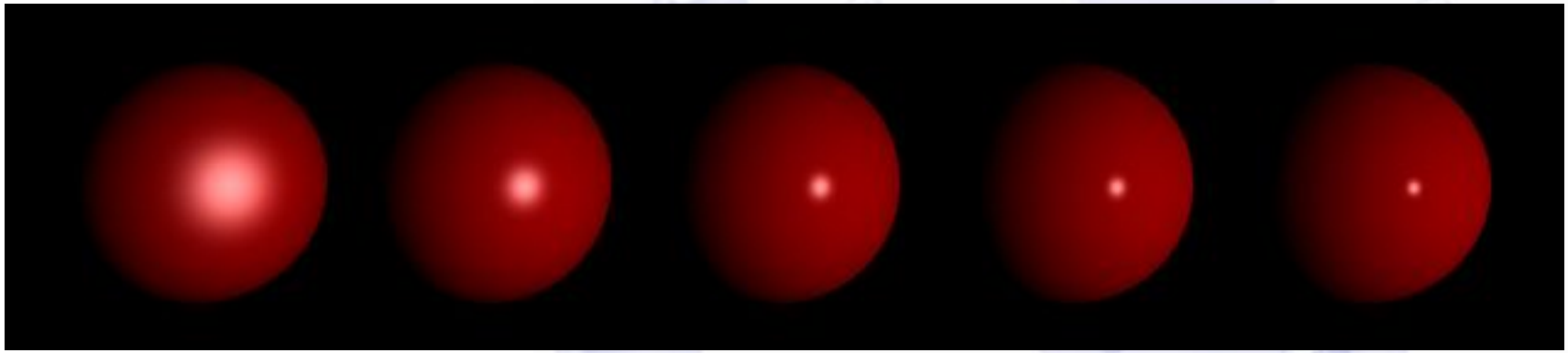
Where v and r (direction of reflection) are unit vectors

specular (k_s) values of 0, 0.25, 0.5, 0.75, 1

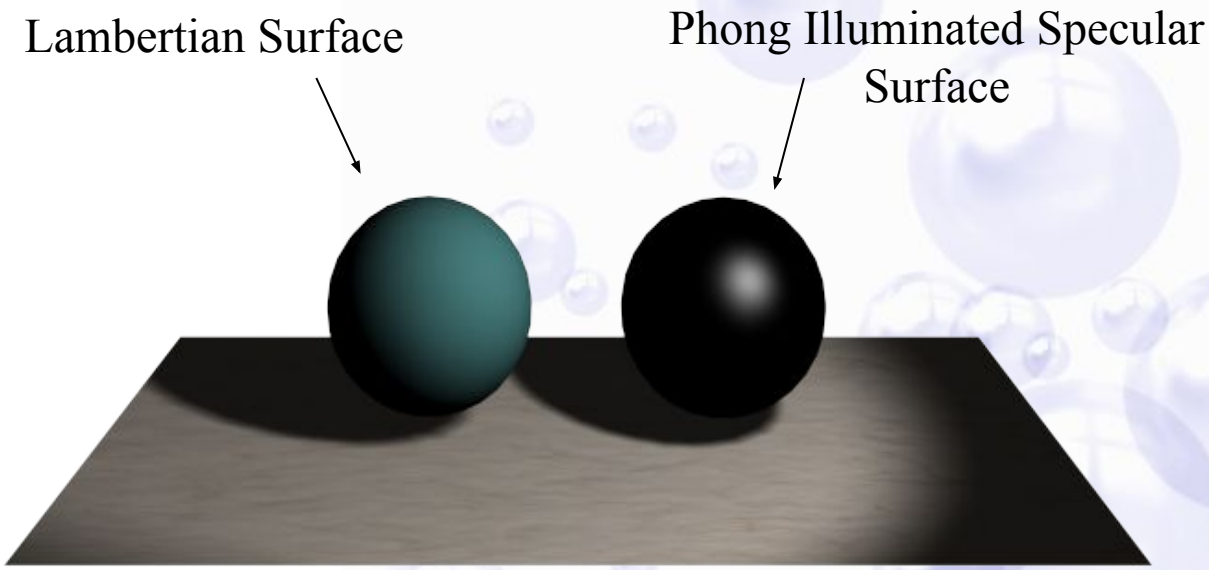
shininess (α) values of 5, 25, 75, 125, 225



Combined with a constant diffuse red component

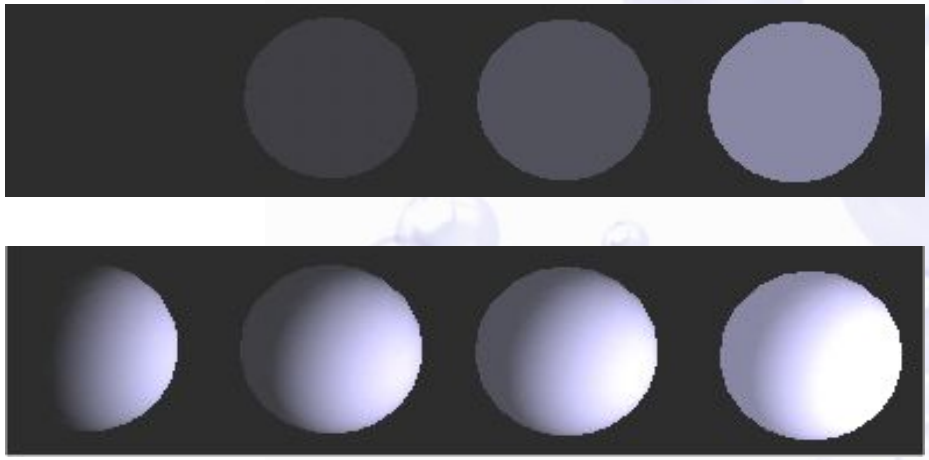


Pure Lambertian vs. Phong



Ambient Light

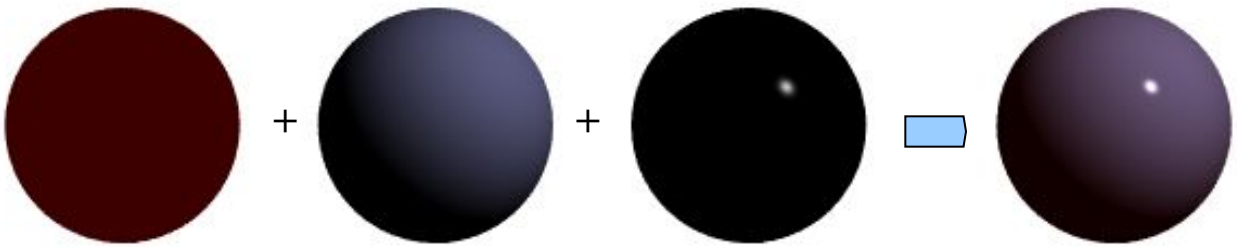
- Light scattered in the scene is modelled using an ambient component – a small level of colour added to all objects in the scene



$$I_a = L \times k_a$$

Putting it all together

- The intensity of light from one point is a sum of the diffuse, specular and ambient components:



Red ambient Bluish diffuse Specular Highlight

$$I = I_a + I_d + I_s$$

OR

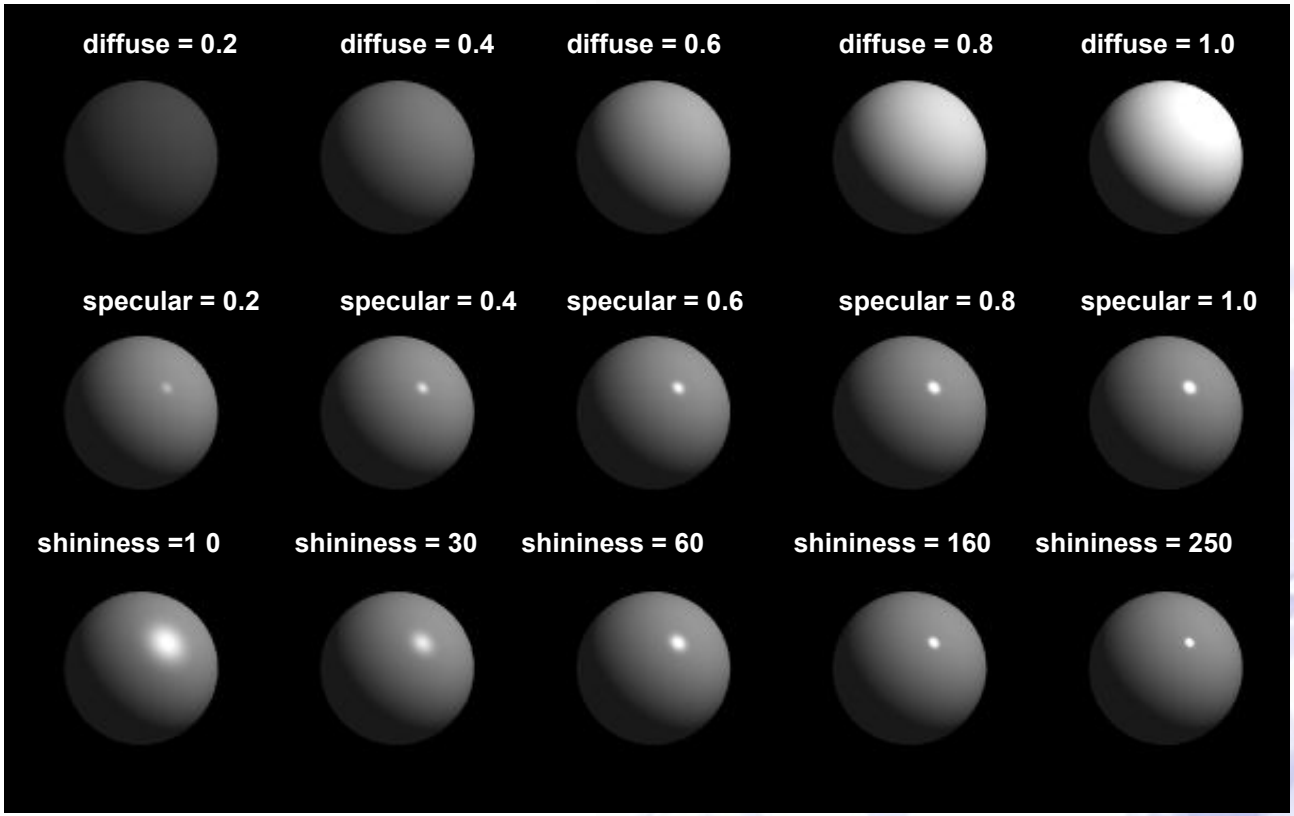
$$I = L \times k_d \times (\mathbf{l} \bullet \mathbf{n}) + L \times k_s \times (\mathbf{r} \bullet \mathbf{v})^{\alpha} + L \times k_a$$

We have just been dealing with single intensities – i.e. greyscale. For a colour solution kd, ka, ks each are a vectors of three components (red, gree, blue). And we need to solve the above equation for each primary colour. i.e.

$$I_{RED} = L_{RED} \times k_{d\,RED} \times (\mathbf{l} \bullet \mathbf{n}) + L_{RED} \times k_{s_{RED}} \times (\mathbf{r} \bullet \mathbf{v})^{\alpha} + L_{RED} \times k_{a_{RED}}$$

$$I_{GREEN} = L_{GREEN} \times k_{d\,GREEN} \times (\mathbf{l} \bullet \mathbf{n}) + L_{GREEN} \times k_{s_{GREEN}} \times (\mathbf{r} \bullet \mathbf{v})^{\alpha} + L_{GREEN} \times k_{a_{GREEN}}$$

$$I_{BLUE} = L_{BLUE} \times k_{d\,BLUE} \times (\mathbf{l} \bullet \mathbf{n}) + L_{BLUE} \times k_{s_{BLUE}} \times (\mathbf{r} \bullet \mathbf{v})^{\alpha} + L_{BLUE} \times k_{aBLUE}$$



specular = 0

shininess = 0

diffuse = 0.5

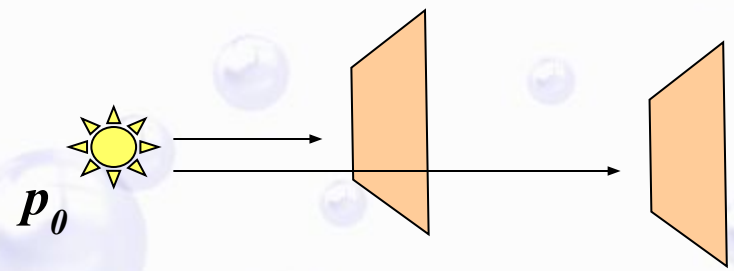
shininess = 120

diffuse = 0.5

specular = 0.5

Light Source Attenuation

- Using the model so far, two parallel planes at different distances from the light source would be rendered exactly the same:
  - Distance from source seems to have no effect



- We need to account for energy transport falling off with distance from source:

$$I = L_a k_a + f_{att} I_d k_d (\overline{N} \bullet \overline{L})$$

- For a point light source the inverse square law (intensity falls off in proportion to the square of distance from source) is a correct model:

$$f_{att} = \frac{1}{d_L^{\,2}} = \frac{1}{|p - p_0|^2}$$

Light Source Attenuation (2)

- However this is not a good model in practice (largely because most objects in the real world are not lit by point sources).
- A better approximation which allows for a richer range of effects is:

A popular model is the Quadratic attenuation model:

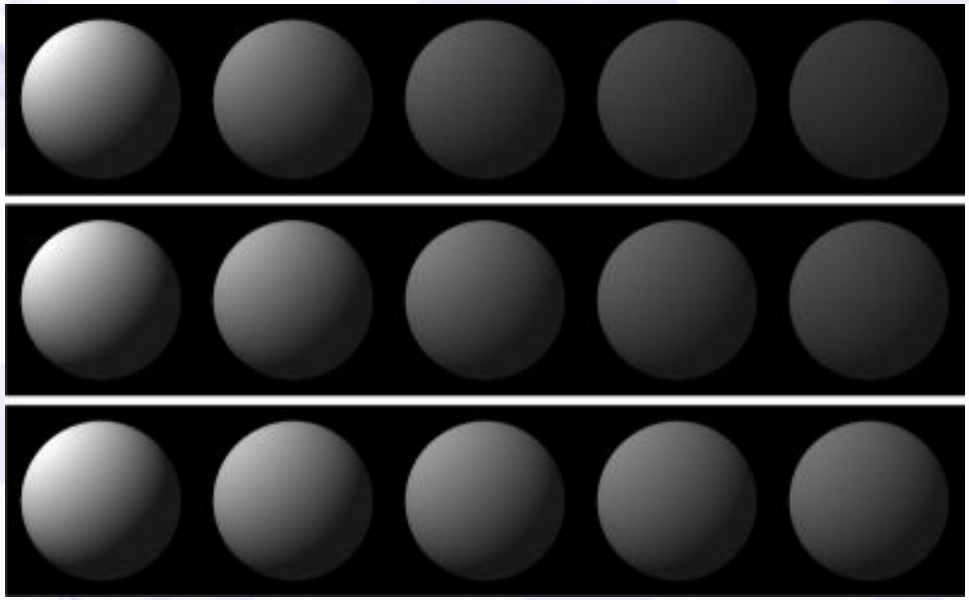
$$f_{att} = \frac{1}{a + bd_L + cd_L^{\ 2}}$$

Spheres at increasing distances from light source

a=b=0; c=1

a=b=0.25; c=0.5

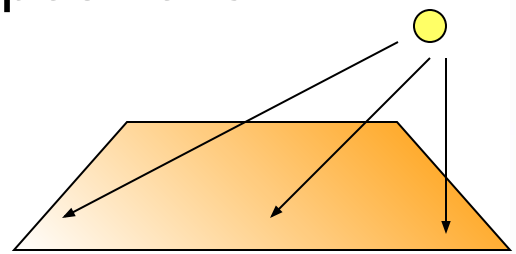
a=0; b=1; c=0



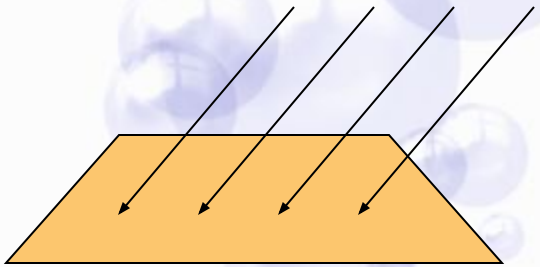
Distant Light Sources

- Shading calculations usually require direction from point on surface to light source, this vector needs to be recomputed at each point
- If light source is distant, the effect is that all beams are approximately parallel
- In lighting calculations we simply replace location of light source with direction of light source. In homogeneous coordinates

positional:

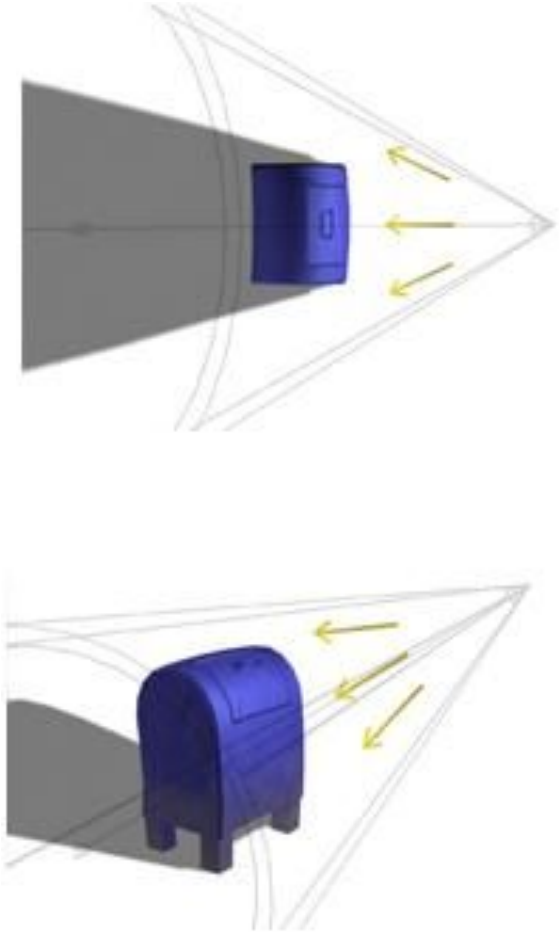


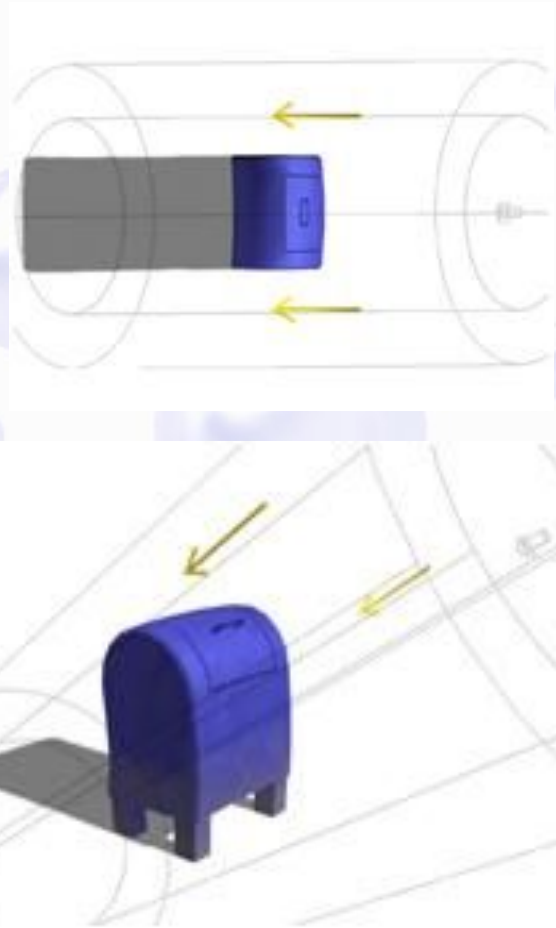
directional:



Positional/Point Light: light source is near – distorts shadow

Directional Light: light source is far away





Positional or Directional Light - Effect on Shadow

Implementation

- Code that implements an illumination is provided here:
  - **http://isg.cs.tcd.ie/dingliaj/3d4/LocalIllumination.cpp**
  - **http://isg.cs.tcd.ie/dingliaj/3d4/localIllumination.h**
- Code is not really optimised for speed.